

HYEONTAE SUNG

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EDUCATION

Korea Advanced Institute of Science & Technology (KAIST) , Republic of Korea <i>Master of Science, School of Electrical Engineering</i> Advisor: Professor Heejin Ahn	Daejeon, South Korea <i>Feb 2023 — Feb 2025</i>
Konkuk University <i>Bachelor of Science, Department of Mechanical and Aerospace Engineering</i> GPA: 4.34/4.5 , Rank: 1/45	Seoul, South Korea <i>Mar 2016 — Feb 2023</i>

RESEARCH INTERESTS

Safety Challenges of Autonomous Systems; Safety-critical, Stochastic, Optimal, and Learning-based Control; Robotics and Autonomous driving

RESEARCH EXPERIENCE

Safe and Intelligent Autonomy (SIA) Lab, Stanford University, Remote July 2025 — Present
Research intern (Advisors: Professors Somil Bansal and Jason J. Choi)

- Safety Filter for High-Dimensional Systems under Uncertainties
 - developed a probabilistic safety filter framework leveraging particle-based reachability value functions
 - enhanced safety of high-dimensional systems under uncertainties without relying on worst-case assumptions
 - validated our method on social navigating problems with chance-constrained model predictive control baseline
 - refined and trained a neural PDE solver to compute the particle value function of Hamilton-Jacobi Reachability

Control and Intelligent Systems Laboratory, KAIST, Republic of Korea Jan 2023 — Present
Research Assistant (Advisor: Professor Heejin Ahn)

- Safe Chance-constrained Model Predictive Control (MPC)
 - designed an MPC framework for autonomous driving under Gaussian mixture model (GMM) uncertainty
 - theoretically guaranteed recursive feasibility under an assumption we proposed
 - proposed a probabilistically recursive feasible MPC without the assumption
 - conducted simulations with the CARLA simulator and the trajectory forecasting model, *Trajectron++*
- Optimal Acceleration Profiles of Multiple Robots
 - designed an acceleration planning system maximizing throughput per unit of time while preventing collision
 - a collaborated project with Samsung Electronics

Systems Control and Multiagent Optimization Research, EPFL, Switzerland Jun — Sep 2024
Research intern (Advisor: Professor Maryam Kamgarpour)

- Real-time Chance-constrained MPC for Robot Trajectory Planning:
 - implemented the safe chance-constrained MPC framework on the robot testbed
 - addressed the sim-to-real gap by using transfer learning methods
 - improved computational time to enable real-time application

In PREPARATION

Safety Filtering for High-Dimensional Stochastic Systems via Particle Reachability Value Functions
In collaboration with Zeyuan Feng, Jason J. Choi, and Somil Bansal (working title)
Link: <https://hyeontaesung.github.io/Stochastic-DeepReach/>

JOURNALS and CONFERENCES

Probabilistic Recursively Feasible Model Predictive Control Under Uncertain Environments

Sung, H., Ham, H., Park, J., Ren, K., & Ahn, H.

Submitted to the 23rd IFAC World Congress link: <https://hyeontaesung.github.io/PRF-MPC/>

Recursively Feasible Chance-constrained Model Predictive Control under Gaussian Mixture Model Uncertainty

Ren, K., Chen, C., Sung, H., Ahn, H., Mitchell, I., & Kamgarpour, M.

IEEE Transactions on Control Systems Technology, 2024 link: <https://ieeexplore.ieee.org/document/10745535>

Safety Filtering Using Sampling-Based Model Predictive Control

Park, J., Sung, H., & Ahn, H.

To appear in International Conference on Robot Intelligence Technology and Applications. Springer, 2025.

Computational Analysis on Safe Chance-constrained Model Predictive Control under Gaussian Mixture Model Uncertainty

Sung, H., Kim, K., & Ahn, H.

Institute of Control, Robotics and Systems (ICROS) 2024 (Best Paper Award, top 2.1%)

Optimal Acceleration Profiles of Multiple Vehicles Using Model Predictive Control

Sung, H., Choi, E., & Ahn, H.

Institute of Control, Robotics and Systems (ICROS) 2024

AWARDS AND SCHOLARSHIPS

EPFL Excellence in Engineering (E3) Summer Fellowship (1.9% over 2060 applicants) 2024
École Polytechnique Fédérale de Lausanne (EPFL) Lausanne, Switzerland

Best Paper Award — 1st Place (8 out of 372, top 2.1%) 2024
Institute of Control, Robotics and Systems (ICROS) 2024 Daejeon, South Korea

National Full Scholarship, KAIST 2023 — 2024
For master's degree Daejeon, South Korea

UAM Vertiport Design Contest—3rd place 2021
Korea Airports Corporation Seoul, South Korea

The 10th EDISON SW Utilization Contest—1st Place 2020
Minister of the Ministry of Science and ICT (Information and Communication Technology) Seoul, South Korea
Optimizing the Main Wing of a High-Altitude Long-Endurance Unmanned Aerial Vehicle Based on Battery Location

Academic Excellence Scholarship for three semesters
Konkuk University Seoul, South Korea

Dean's list for one semester Fall 2020
Konkuk University Seoul, South Korea

SKILLS

- **Relevant Coursework:** Optimal Control Theory (A+), Convex Optimization (A+), Linear Systems (A)
- **Programming:** MATLAB, Python, C/C++.
- **Software:** CARLA Simulator, ROS 2, Inventor, Fusion 360, CATIA.
- **Hardware:** OptiTrack, NVIDIA Jetson Platforms (Jetbot - Mobile Robot, JetRover - Mobile Manipulator)

TEACHING EXPERIENCE

Teaching Assistant

- AI for Robot Intelligence: Perception, Planning and Control Fall 2025
- Signal and Systems Spring 2025
- Electronics Design Lab. <Robotic Manipulator > Fall 2024
- Introduction to Optimization Techniques Fall 2023

Research Mentorship

- Undergraduate Research: Time-dependent Shortest Path Algorithm Winter 2023

ADDITIONAL ACTIVITIES

Reviewer: IEEE Transactions on Control Systems Technology, IROS 2025 2024

Student Lab Representative Dec 2023 — Jun 2024

Exchange Student in Mechanical Engineering at California State University, Sacramento Sep — Dec 2021

Military Service: Served at Military Police Special Duty Team (SDT) as a squad leader Jun 2017 — Feb 2019